Integrated Thermally Tuned Mach-Zehnder Interferometer in Z-Cut Lithium Niobate Thin Film

Guanyu Chen, Hong-Lin Lin, Jun Da Ng, and Aaron J. Danner, *Senior Member, IEEE*

***Abstract*— We propose and demonstrate a type of highly efficient tunable Mach-Zehnder interferometer (MZI) in Z-cut lithium niobate thin film (TFLN) based on the thermo-optical effect. By placing the electrode directly above the TFLN waveguide, a more uniform and higher heat distribution was obtained, compared with the conventional horizontal structure. Both the principle and design strategies of the proposed scheme are described. The process flow of the device fabrication is demonstrated in Z-cut TFLN. The measured static results accord well with the simulations. About 0.121 nm/V tuning efficiency is observed, which is nearly double the tuning efficiency based on the electro-optic effect, and may prove useful when high switching speed is not the primary consideration. The presented thermo-optic tuning scheme can be extended to other scenarios easily and may satisfy many different application requirements in integrated lithium niobate photonics (LNPs).**

***Index Terms*— Lithium compounds, photothermal effects, interferometers, integrated optics, nanophotonics, optical waveguides.**

I. Introduction

LITHIUM niobate (LN) photonics has experienced significant developed in recent decades due to its excellent performance [1]–[7]. Compared to silicon photonics, there are several unique advantages for LN, such as ultra-high second order susceptibilities (r_{33} ~30 pm/V), wide transparency and stable physical and chemical characteristics [1], [2]. For decades, devices fabricated in single crystal bulk LN have primarily made use of the ion diffusion method to great success, and the lithium niobate electro-optic modulator is ubiquitous in fiber-optic communication systems [1], [2]. However, devices based on bulk LN usually feature large device sizes and large waveguide bending loss due to its weak refractive index contrast [1], [2]. The recent development of high-quality LN on oxide (LNOI) thin film [3]–[7] makes integrated LN photonics (LNPs) possible. Various types of high-performance LN thin film (TFLN) based devices have been demonstrated [3]–[10], which together have boosted the development of LNPs in general and is paving the way to large scale integration. Among integration application scenarios, wavelength tunability is an important function, as it can play a role in wavelength stabilization [6], tunable interleavers [8], and switching [11], for instance. In most reported results, index tunability is realized based on the electro-optic (Pockels) effect [5], [6], [8]. However, this usually requires either a high voltage or long device length [5], [6], [8], as r_{33} is not high enough for the accumulation of significant refractive index change; this physical limit impedes chip-scale dense integration of optical devices. The thermo-optic (TO) effect, on the other hand, allows changing a material's refractive index using heating and has certain advantages [12]. Although TO-based devices have been widely demonstrated in other material platforms [12]–[14], it has seldom been explored in TFLN. Additionally, most demonstrated electro-optic TFLN-based devices are fabricated on X-cut LN with electrodes typically located laterally with respect to a given waveguide [3]–[6], [8]. Such a configuration is not useful for Z-cut LN as the crystal orientation is perpendicular to the direction of required applied field [1]. Higher performance is expected in electro-optic devices by placing electrodes along the vertical direction of Z-cut LN, as the electric field has greater overlap with the propagating optical mode relative to the situation with lateral electrodes in the X-cut case [1]. The thermo-optic effect should also benefit from such a vertical structure as the heat transferred into the underlying waveguide will be more uniform.

Therefore, we have demonstrated a type of highly efficient thermally tuned Mach-Zehnder interferometer in Z-cut LNTF with a vertical electrode structure. The principle and design methodology of the proposed device is described. The electrode is placed above the waveguide in our scheme. Thus, a more uniform and strong heat distribution can be realized. The device has been fabricated and the process flow is described in detail. Both the static and dynamic performance of the proposed structure are characterized. About 1.3 nm free spectral range (FSR) is observed, which accords well with simulation results. The measured thermal tuning efficiency is about 0.121 nm/V based on an aluminum (Al) electrode, which is nearly double that of schemes based on the electro-optic effect. We believe the proposed highly efficiency thermal tuning Mach-Zehnder interferometer with a vertical structure can fulfill many different application scenarios where the rapid switching speeds of the Pockels effect are not required, but where significant index tuning can be achieved. The use of

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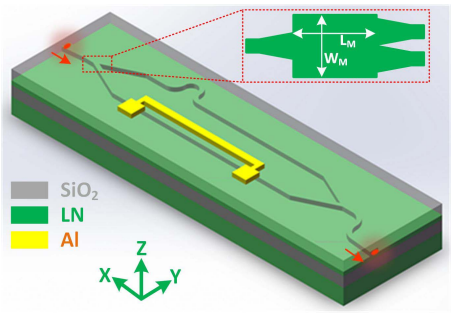

Fig. 1. Schematic structure of the proposed device.

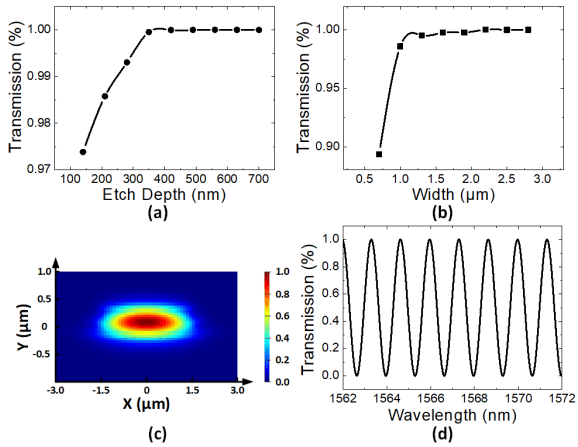

Fig. 2. Simulated transmission results with respect to (a) etch depth and (b) waveguide width. (c) Simulated mode distribution of the waveguide. (d) Simulated transmission spectrum of the asymmetric Mach-Zehnder interferometer.

both physical effects in tandem may lead to ultimately greater device utility in integrated lithium niobate photonics going forward.

II. Principle and Design

The schematic structure of the proposed thermal tuning TFLN Mach-Zehnder interferometer is shown in Fig. 1. The design is based on Z-cut LNOI structure with 700 nm top LN and 2 μm underlying SiO_2. The refractive indices of LN and SiO_2 are 2.3 and 1.44, respectively, which form a large index difference and thus can realize compact integrated structure compared with typical ion diffusion based bulk device [1]. For a Z-cut LN based electro-optic devices, r_{33} will only maximize its effect when light is polarized vertically [1]. Therefore, the transverse magnetic (TM) polarized mode is chosen in our thermo-optic design and experiment for better ultimate compatibility with other application scenarios, even though the electro-optic effect is not used here.

The basic waveguide geometry is designed based on the 3D finite difference time domain (FDTD) method after considering both the transmission loss and fabrication technology. Although a strip waveguide would have better mode confinement, it would be hard to realize in experiment due to the troublesome but well-known etching properties of LN [15]. According to our simulation shown in Fig. 2(a), the improvement of transmission loss is relatively small when the etch depth is more than 400 nm for a 700 nm thick LN layer (etch ratio is about 57%). Therefore, a ridge waveguide with 400 nm etch depth is used for our design. As for the waveguide width, acceptable transmission loss can be obtained when it exceeds 1 μm, as shown in Fig. 2(b). However, too great a width will activate higher modes, which will degrade device performance at the split. Therefore, 2 μm is chosen as our waveguide width. The simulated TM mode profile for the waveguide with 2 μm width and 400 nm etch depth is shown in Fig. 2(c). Two such parallel ridge waveguides are connected through two 1 $\times$ 2 multimode interferometers (MMIs) for dividing and combining light at the input and output, respectively. For facilitating measurement, the width of the waveguide at each end tapers to 10 μm adiabatically. The length (L_M) and width (W_M) of the multimode region for the 1 $\times$ 2 MMI are 15.5 and 5.0 μm, respectively. The Mach-Zehnder interferometer is designed as an asymmetric structure with about 800 μm ΔL difference, which will result in a nominal 1.3 nm free spectra range (FSR), as shown in the simulated results in Fig. 2(d).

Thermal tuning is widely used in different material systems [12]–[14]. Its principle is based on the change of refractive index with respect to temperature variation, which is usually referred to as the thermo-optic effect. The refractive index variation is related to phase change. If such a phase difference is applied to one of the two MZI arms, the operating wavelength will shift due to the interference. The theoretical wavelength change $\Delta\lambda$ with respect to temperature variation ΔT is shown in equation (1), where λ_0 is the operating wavelength, n_g is the group index of LN, and $\frac{\Delta n_e}{\Delta T}$ represents the thermo-optic coefficient of LN. The TO coefficients of LN and SiO_2 are about 3.34×10^{-5} and $0.95 \times 10^{-5} K^{-1}$, respectively.

$$\Delta\lambda = \Delta T \cdot \frac{\lambda_0}{n_g}\left(\frac{\Delta n_e}{\Delta T}\right) \tag{1}$$

It's worth noting that there will be a more uniform and greater distribution of heat when the resistive metal electrode is placed above the waveguide compared with the case of putting the electrode beside the waveguide. To understand this more easily, we have performed thermal-optical simulations based on the finite element method. Figures 3(a) and (b) are the simulated thermal distributions for the two cases of electrodes above (vertical structure) and beside (horizontal structure) the waveguide, respectively. We choose thermal conductivities of the LN, SiO_2 and Al electrode as 5.6, 1.3 and 205 W/(m·K) in our simulation, respectively. In the vertical case, there is a 1 μm thick SiO_2 between the top electrode and the LN waveguide to prevent optical loss due to the metal. The distance between the horizontal electrode and waveguide is also set as 1 μm for a fair comparison. All the other parameters are set as the same during the simulation, except the electrode position. It can be observed from Fig. 3 that the temperature distribution of the vertical structure is more uniform and it's at least 6 K higher compared with the horizontal case, which indicates its superiority.

III. Device Fabrication

The detailed process flow of the fabrication procedure is shown in Fig. 4. The fabrication of the device starts from

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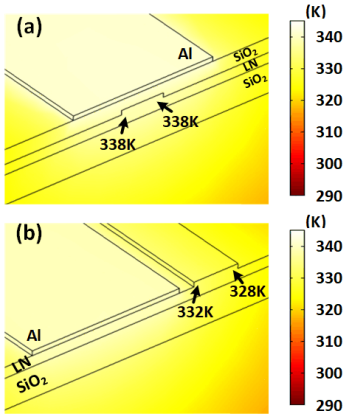

Fig. 3. Simulated thermal distributions when place metal electrodes (a) above and (b) aside the waveguides.

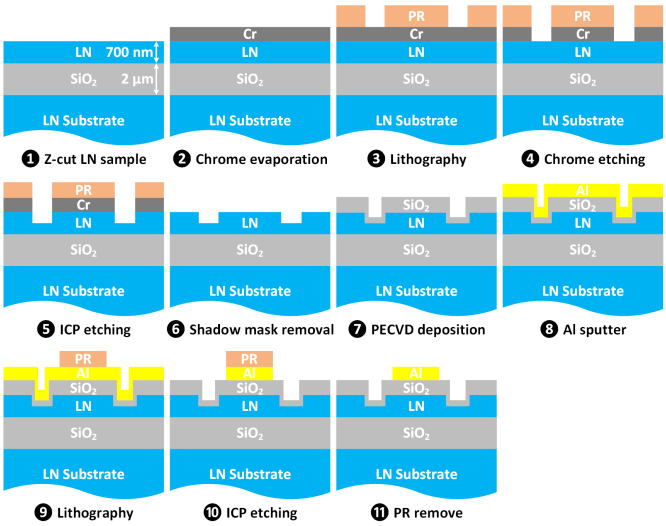

Fig. 4. Process flow of the device fabrication.

a bare Z-cut TFLN wafer with a 700 nm top LN layer. A layer of 2 μm silicon oxide has been inserted between the top layer and LN substrate. After a standard cleaning step, a 50 nm thin layer of chrome (Cr) is first deposited above the surface using electron beam (e-beam) evaporation, which is used for the facilitation of lithography as the sample is transparent. After spin-coating ma-N 1410 negative photoresist (PR), the optical pattern is transferred onto it through a DWL 66+ laser lithography system. Both the photoresist and Cr serve as a shadow mask for subsequent inductively coupled plasma (ICP) dry etching after development and Cr wet etch. The ICP etching is based on Oxford PlasmaPro 100c Cobra with only argon gas [15]. The bias power of the ICP etching is around 320 W. Here the etch rate for LN is about 25 nm/min, with around ~1:1 selectivity for photoresist. After dry etching, the sample is put into hot N-methyl-2-pyrrolidone with assistance of ultra-sonic cleaning for the removing of photoresist. The thin layer of Cr is stripped by Cr etchant. Then the sample is cleaned using 10% hydrochloric acid (HCl: $H_2O = 1 : 10$) for 60 s. After that, about 1 μm silicon oxide is deposited above it based on plasma enhanced chemical vapor deposition (PECVD). A layer of 300 nm Al is then

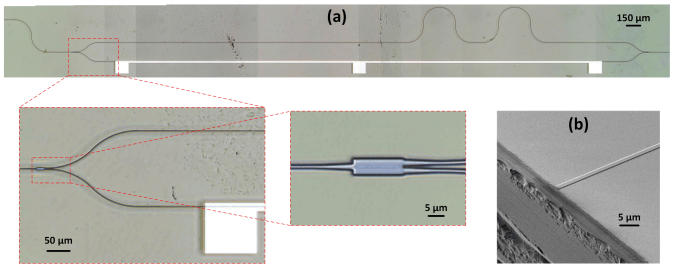

Fig. 5. (a) Microscope images of the fabricated device. (b) SEM image of the edge coupler facet.

deposited using a sputtering system. A similar lithography step using ma-N 1410 negative PR is performed to define the electrode pattern. The redundant Al on the sample is removed using chlorine based ICP. The photoresist is removed using hot N-methyl-2-pyrrolidone which completes the fabrication procedure.

Figure 5(a) shows microscope images of the fabricated device. At one end, the waveguide at one end has an S-bend to avoid direct coupling of light through free space during measurement. All the bending radii are 150 μm. The zoomed-in pictures show the detail of the MMI and the electrode pad. The light is coupled in and out from the chip through inverse tapered couplers, which are diced with a precision blade (Accretech). A two-step dicing method is used here, where a shallow trench is first sawed by a relative slow and grinding-like step, and then cut past the substrate with greater depth. The scanning electron microscope (SEM) image of the waveguide after dicing is shown in Fig. 5(b).

IV. Experiment Results

The device is characterized based on the setup of Fig. 6. The output of an amplified spontaneous emission (ASE) light source (ALS-CL-15-B-FA) is collected by a single mode fiber. As mentioned before, TM light is used during the measurement. Thus, a polarization controller (PC) is placed in the front of coupling setup for tuning of polarization states. The input light is coupled into the chip through a three-dimensional coupling setup, with the assistance of a near-infrared (NIR) camera. The bias voltage of the device is applied through a source meter (Keithley 2400). The output of the device under test (DUT) is injected into an optical spectrum analyzer (OSA) for analysis.

The measured results are shown in Figure 7. Firstly, the transmission spectrum range from 1530 nm to 1600 nm of the device without bias voltage is measured, with results shown in Fig. 7(a). Figure 7(b) is the zoomed-in spectrum from 1550 nm to 1560 nm. A nearly flat spectrum over a wide wavelength range can be observed. The FSR is about 1.3 nm, which accords well with the simulation results. The measured average extinction ratio (ER) is above 10 dB. Then the dynamic behavior of the device is characterized, with measured results shown in Fig. 7(c) and (d). Fig. 7(c) is the measured spectrum under different bias voltages from 1550 to 1560 nm wavelength. Fig. 7(d) is the zoom in result. It can be observed from Fig. 7(c) and (d) that blue shift of the operating wavelength is about 1.21 nm when the bias voltage increases

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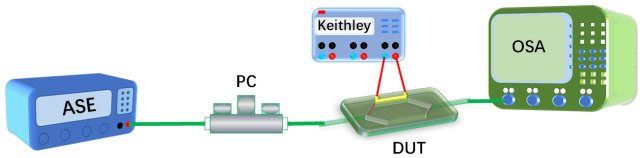

Fig. 6. Measurement setup.

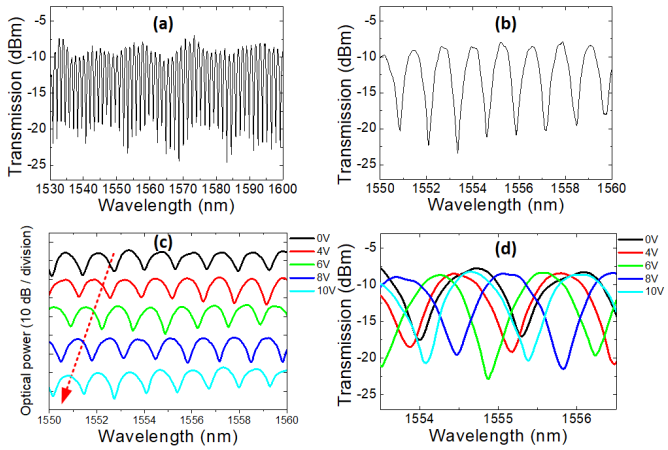

Fig. 7. Measured transmission spectra of the device under zero bias (a) over a wide wavelength range and (b) zoomed-in case. Measured transmission spectra under different bias voltages (c) over a wide wavelength range and (d) zoomed-in results.

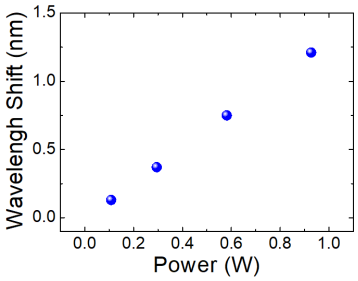

Fig. 8. Measured wavelength shift with respect to applied power.

from 0 V to 10 V, which corresponds to a 0.121 nm/V tuning efficiency.

The applied power with respect to wavelength shift is shown in Fig. 8. The calculated tuning efficiency per power is calculated as 1.32 pm/mW. Compared with electro-optic effect based tuning methods [8], our thermo-optic tuning structure gives nearly double the resonant wavelength shift with the same applied voltage in an MZI of similar length (the comparison is performed by normalizing to the appropriate electrode length in reference [8] and our design). Here the total resistance of the 4500 μm long Al electrode is derived from the IV curve as around 100 Ω. By using electrode material with higher resistivity and air trench structure [10], [16], the tuning efficiency can be further improved. The somewhat lower thermal tuning efficiency compared with other material systems [13], [14] can be attributed to the lower thermal conductivity of LN. The response time of the proposed device is around a hundred microseconds (μs). The lower thermal conductivity of LN compared with other material systems, such as silicon, will cause a bit slower response time. It's worth noting that the TO effect itself is polarization independent. By engineering the structure of relevant passive components, polarization-independent TO-based devices are possible.

V. Conclusion

In conclusion, we have proposed and fabricated a highly efficient tunable Mach-Zehnder interferometer in a Z-cut LNTF platform based on the thermo-optic effect. By placing the electrode above the TFNL waveguide, the applied heat can be transferred into the target area directly, thus resulting in highly efficient tuning. The design details and process flow of the proposed device have been described. Both the static and dynamic performance of the device have been characterized. Nearly 0.121 nm/V thermal tuning efficiency is measured based on our proposed structure. Such a value can be further improved by replacing this metal with a material of greater resistance or by using air trench structure. We believe the proposed structure in this manuscript can be extended to various application scenarios and aid the development of integrated lithium niobate photonics.

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